UNIT 1 · TOPIC 3(C)

Sowing Equipment

Syllabus topic: Equipment for land development, tillage, sowing, planting and transplanting

Methods of sowing

Method	What it does
Broadcasting	Scatters seed over the surface.
Drilling	Places seed randomly within rows.
Precision planting	Places single seeds at uniform intervals.
Dibbling	Places seeds in holes.
Hill dropping	Drops groups of seeds at intervals.
Check-row planting	Places hills in both directions, so the crop can be cross-cultivated.
Transplanting	Plants seedlings instead of seed.

Seed drill — components and functions

Components, in the order the seed moves through them:

1. Frame
2. Hopper
3. Seed metering mechanism
4. Seed tube
5. Furrow opener
6. Covering device

Transport/drive wheels and a ground-drive transmission power the metering.

What the drill does, in sequence:

1. Meters the seed.
2. Opens the furrow.
3. Places seed at a uniform depth.
4. Covers and compacts the seed.

Seed metering mechanisms

Type	Working	Use
Fluted roller (fluted wheel)	Adjustable exposed length of flutes carries seed; positive, force-feed	Most common on Indian seed drills (wheat etc.)
Internal double-run	Double-faced wheel (large/small side) inside a casing	Grain drills; large + small seeds
Cup feed	Cups on a vertical disc pick seed from the hopper base	UK-type drills, paddy
Cell/plate type — edge drop, flat drop, hill drop	Cells on a rotating plate hold single seeds or groups	Row-crop planters (maize, cotton)
Vacuum/pneumatic disc	Suction holds seed on a perforated disc	Precision planters — vegetables, maize
Inclined plate	Cells on an inclined rotating plate; gentle	Groundnut (minimises seed damage)
Stationary opening + agitator	Adjustable orifice with an agitator	Broadcasters, vegetable seeders

- Fluted-wheel metering is **more positive/uniform than double-run** (Kepner §11.6). Change the seed rate by shifting the flute exposure.
- Fluted rollers cause the most damage to large, fragile seeds; inclined-plate and pneumatic types minimise damage.

Furrow openers (Kepner §11.7)

Opener	Best for
Shoe (fixed)	Trashy, hard soils — the Indian standard.
Shovel / hoe (fixed)	Stony soils.
Full runner (fixed)	Mellow, trash-free soils; maintains depth — maize planters.
Stub runner (fixed)	—
Single disc (rotating)	Cuts trash — minimum-till fields.
Double disc (rotating)	Precise seed placement; sticky soils that will not scour fixed openers.

Covering devices: chains, press wheels, paired covering discs.

Special drills

- **Seed-cum-fertilizer drill:** separate hoppers and metering for seed and fertilizer. It places fertilizer **5 cm below and 5 cm beside** the seed (band placement).
- **Zero-till drill:** inverted-T openers sow straight into unploughed stubble. It saves **~60–80%** of the energy of conventional sowing of wheat after rice.
- **Happy Seeder:** a zero-till drill + flail straw-management rotor. It sows wheat into combine-harvested paddy residue (in-situ residue management).
- **Super Seeder:** a rotavator ploughs the residue in and the machine sows in the same pass.

- Related conservation machines: strip-till drill, Turbo seeder, roto-till drill.

Calibrating a seed drill

1. Jack up the drive wheel.
2. Mark the wheel and rotate it **N** revolutions, equal to a known area: **Area covered = π × D × N × W × n** (D = wheel diameter, W = row spacing, n = number of rows).
3. Collect the seed from each tube and weigh it.
4. Calculate: **Seed rate (kg/ha) = collected mass × 10,000 ÷ area**.

The full calibration procedure is in Unit 3; the definition is also asked in Unit 1.

Practice MCQs

Q23. Which seed-metering mechanism is most common on Indian seed drills?

(a) Cup feed
(b) Internal double run
(c) Fluted roller
(d) Vacuum disc

Q24. Compared with the internal double-run feed, the fluted-wheel feed is:

(a) Less positive in metering
(b) More positive/uniform
(c) Identical
(d) Unsuitable for wheat

Q25. Which furrow opener is best for trash-covered, minimum-till fields?

(a) Shoe
(b) Full runner
(c) Single-disc
(d) Shovel

Q26. In a seed-cum-fertilizer drill, fertilizer is placed relative to seed at about:

(a) Same level, mixed
(b) 5 cm below and 5 cm to the side
(c) 10 cm above
(d) 15 cm below only

Q27. Happy Seeder is used for:

(a) Transplanting rice
(b) Direct drilling of wheat into combine-harvested paddy residue
(c) Planting sugarcane setts
(d) Sowing on ridges

Answers

Q23 — (c) Fluted roller

Q24 — (b) More positive/uniform — *Kepner §11.6 — the fluted-wheel feed is "more positive in its metering action than the double run."*

Q25 — (c) Single-disc

Q26 — (b) 5 cm below and 5 cm to the side

Q27 — (b) Direct drilling of wheat into combine-harvested paddy residue